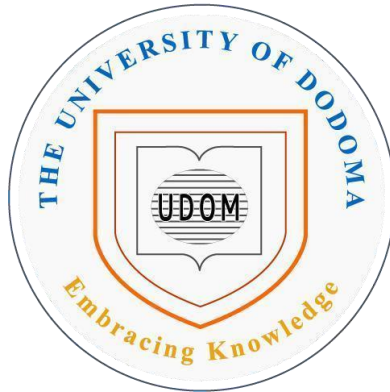


THE UNIVERSITY OF DODOMA



COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(CSE)

COURSE NAME: ROBOTICS AND AUTOMATION

COURSE CODE: CT 322

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1. Introduction

Industrial robots have become an important part of modern manufacturing due to their speed, accuracy, and repeatability. Computer-Aided Design (CAD) models enable engineers to inspect robot structures before manufacturing and allow robot simulations using MATLAB Simulink and Simscape Multibody.

This project analyses an EPSON T3 SCARA robot exported from SolidWorks. The robot structure, links, joints, degrees of freedom, motion characteristics, and workspace are identified and explained.

2. Selected Robot ZIP File

Item	Description
ZIP File	assembly 2009010 epson t3
Assembly File	assembly 2009010 epson t3.SLDASM
Robot Type	SCARA Robot
CAD Software	SolidWorks
Simulation Software	MATLAB Simulink + Simscape Multibody

The uploaded Simscape XML identifies the root assembly as "**assembly 2009010 epson t3**" and references the SolidWorks assembly file.

Figure 1

Insert your uploaded robot image here.

Caption:

Figure 1: Complete EPSON T3 SCARA Robot Assembly

3. CAD Assembly Inspection

The SolidWorks model contains the major mechanical components required for robot operation.

Main Parts

Part	Function
Base	Supports the robot
Connecting Arm	Connects base to second arm
Arm	Carries end-effector
Shaft	Vertical movement
Stopper	Motion limitation
Cable	Electrical connection

The XML export lists these components as part of the assembly.

4. Robot Type and Degrees of Freedom

The robot is classified as a SCARA (Selective Compliance Assembly Robot Arm).

Characteristics

- Two rotational joints
- One vertical linear movement
- End-effector movement

Estimated Degrees of Freedom

4 DOF

Applications

- Pick and Place
- Packaging
- Assembly
- Electronics Manufacturing

5. Link and Joint Identification

Joint	Connected Parts	Joint Type	Motion
J1	Base – Link 1	Revolute	Rotation
J2	Link 1 – Link 2	Revolute	Rotation
J3	Link 2 – Z Axis	Prismatic	Vertical
J4	Z Axis – End Effector	Revolute	Tool Rotation

Figure 2

Insert labelled robot image showing

- Base
- Link 1
- Link 2
- Z-axis

- End-effector
-

6. Robot Motion Reference Points

Position	Description
Home	Initial position
Object	Object detected
Pick	Gripper lowered
Place	Destination
End	Robot returns home

7. Basic Robot Motion Description

Step 1

Robot begins at Home Position.

Step 2

Joint 1 rotates toward the object.

Step 3

Joint 2 extends the arm.

Step 4

Z-axis lowers.

Step 5

Gripper picks the object.

Step 6

Robot lifts the object.

Step 7

Robot rotates toward destination.

Step 8

Robot lowers the object.

Step 9

Gripper releases.

Step 10

Robot returns home.

Motion Sequence Table

Time(s)	J1	J2	J3	J4
0.0	0°	0°	Up	Open
0.5	20°	15°	Up	Open
1.0	35°	25°	Down	Close
1.5	-20°	30°	Up	Close
2.0	0°	0°	Up	Open

8. MATLAB / Simulink Preparation

The robot was exported from SolidWorks using Simscape Multibody Link.

Simulation preparation included

- Import XML
- Detect robot bodies
- Detect joints
- Connect World Frame
- Test robot movement

The XML indicates it was created using Simscape Multibody Link from SolidWorks.

Motion Objective

Move object from Home Position to Pick Position and finally Place Position.

Motion Duration

2 Seconds

Insert Motion Sequence Screenshots

10. Workspace Observation

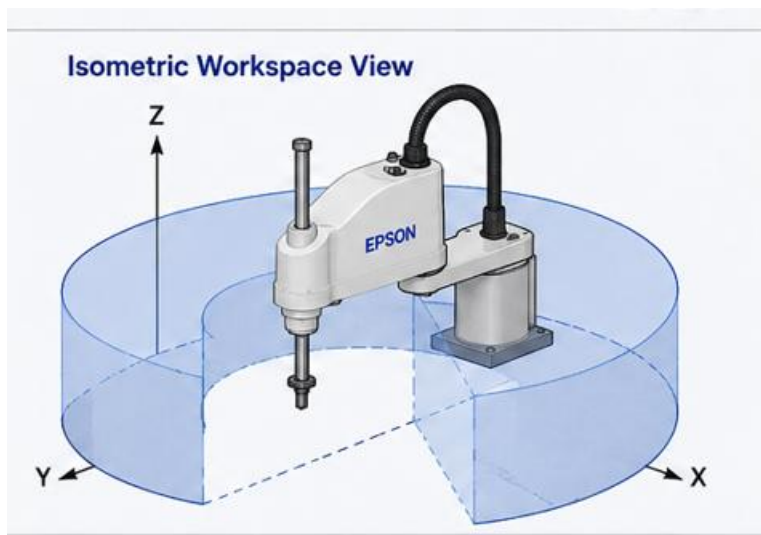
The robot has a semicircular working envelope.

Maximum reach is achieved when both arm links are fully extended.

Limitations include

- Limited vertical travel
- Limited payload
- Fixed installation position

Figure 5



11. Practical Applications

- Electronics assembly
 - PCB manufacturing
 - Packaging
 - Medical laboratories
 - Pharmaceutical automation
 - CNC loading
 - Inspection
-

Limitations

- Limited payload
 - Limited reach
 - Fixed workspace
 - High maintenance cost
 - Requires calibration
-

12. Conclusion

The EPSON T3 SCARA robot demonstrates excellent performance for high-speed industrial automation tasks. Its SCARA configuration provides fast and accurate planar motion while maintaining good positional accuracy. The CAD model successfully illustrates the robot structure and can be prepared for simulation using MATLAB Simulink and Simscape Multibody.

References

1. MATLAB Simscape Multibody Documentation.
 2. SolidWorks User Guide.
 3. Robotics: Modelling, Planning and Control.
 4. Peter Corke, Robotics Toolbox.
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Appendix

Include

- XML Export
- SolidWorks Screenshots
- MATLAB Screenshots
- Joint Tables
- Additional Figures